Shahjalal University of Science and Technology

Department of Electrical and Electronic Engineering

4th Year 1st Semester Final Examination-2023

Course Code: EEE 455 Session: 2019-20

Course Title: VLSI I

Total Marks: 100 **Credit: 3.0** **Total Time: 3 Hours**

Part - A

(Answer any two of the following questions)

1. (a) Briefly discuss the chemistries involved in in [*sic*: should be "involved in"] etching. Also explain the effect of adding H_2 and O_2 on etching rate. 6

   (b) What is RCA clean? What are the steps? What contaminants can be removed in this process? 6

   (c) Define photolithography. briefly describe the major steps involved in photolithography. 8

   (d) Explain the following non-idealities of MOSFET- (i) Drain induced barrier lowering, (ii) subthreshold leakage. What are the effects of temperature on these non-idealities? 5

2. (a) Find the equations and I-V characteristics curve of MOSFET current using long-channel model. 8

   (b) Find propagation and contamination delay (for both rising and falling) of 3-input NAND gate while it is driving 4 identical NAND gates. Consider that the gate has equal pull-up and pull-down resistance. 7

   (c) State Moore's law. Discuss the problems that arise due to MOSFET scaling. 6

   (d) Explain the principle of cascode voltage switch logic. 4

3. (a) Using I-V characteristics curve of MOSFET, find DC characteristics of CMOS inverter. 7

   (b) Find minimum delay for path $A \rightarrow B$ in the logic circuit of Figure 3.1. Also find the gate size to achieve this delay. 6

   (c) Figure 3.2 shows a portion of a decoder circuit. The inverter at input drives 8 identical 4-input NAND gate with input capacitance y each. The output of the decoder (NAND gate) is connected to a 32-bit word of register and each register bit presents a load of 3-unit sized transistor. Find best number stages (how many inverters should be added between NAND and load?), stage effort, path delay, NAND gate size. 7

   (d) Explain input ordering delay effect. 5

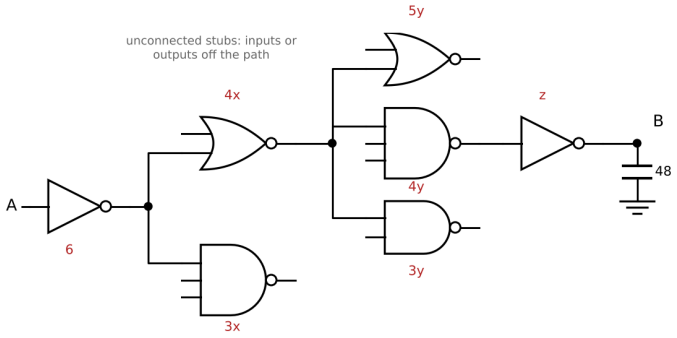


Figure 3.1 of the paper (redrawn)

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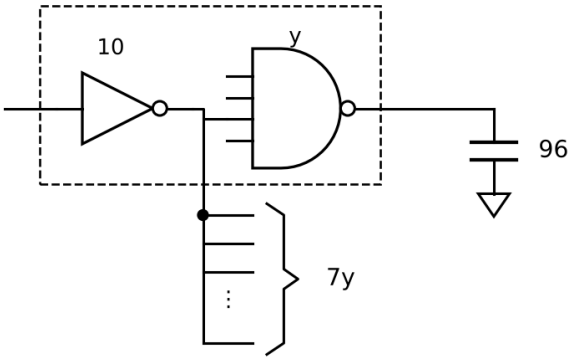


Figure 3.2 of the paper (redrawn from the hand-drawn sketch on the paper)

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Part - B

(Answer any two of the following questions)

4. (a) What is scaling? Explain how different parameters of transistors are scaled? 5

   (b) Define glitch? How does it increase power consumption? How it can be reduced? 6

   (c) You are synthesizing a chip composed of random logic with an average activity factor of 0.1. You are using a standard cell process with an average switching capacitance of 450 pF/mm^2. Estimate the dynamic power consumption of your chip if it has an area of 70 mm^2 and runs at 450 MHz at $V_{DD} = 0.9$ V. 4

   (d) Define crosstalk. Show how crosstalk induces noise in different cases. How can crosstalk be reduced? 7

   (e) Define sheet resistance. How does interconnect affect the performance of a system? 3

5. (a) What is min-delay constraint? Find min-delay for a system sequenced with flip-flop. If min-delay occurs, what is the solution? 6

   (b) The Itanium 2 ALU self-bypass path is shown in Figure 5.1, and delays associated with each element are given in Table 5.1. Calculate the minimum cycle time T_c at which ALU self-bypass path will operate correctly. The earliest input to the late bypass multiplexer is imm value coming from another flip-flop. Will this path experience any hold-time failures? 4

   (c) Design a CMOS logic circuit to compute $F = AB + CD$ using (i) only NAND gates, (ii) compound gate. Choose the transistor width to achieve unit path resistance. If each input can present a maximum of 20 λ of transistor width, output must drive a load equivalent to 100 λ of transistor width, calculate the minimum delay to compute F. Which one is faster? Comment on your answer. 9

   (d) Define stack effect. Find subthreshold leakage current for two series OFF transistor. 6

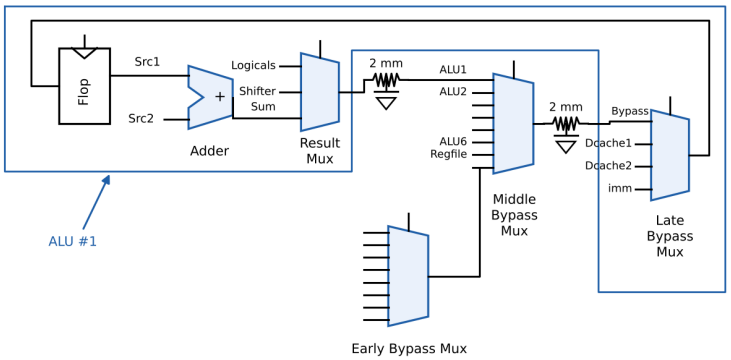


Figure 5.1 of the paper: the Itanium 2 ALU self-bypass path (redrawn)

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Elements	Propagation delay	Contamination delay
Adder	590 ps	100 ps
Result Mux	60 ps	35 ps
Early Bypass Mux	110 ps	95 ps
Middle Bypass Mux	80 ps	55 ps
Late Bypass Mux	70 ps	45 ps
2-mm Wire	100 ps	65 ps

Table 1: Combinational Logic Delays (Table 5.1 of the paper)

6. (a) Explain the principle of dynamic logic circuit. 6

   (b) Draw the stick diagram of the following Boolean function: $f = (A+B).(C+D)$ with their symbolic diagram and truth table. 6

   (c) Draw logic diagram of a SRAM cell and explain its different modes of operation. 6

   (d) Design and implement a pseudo-nMOS ROM to store the data shown in Figure 6.1. Also draw the dot diagram representation. 7

word0: 010101

word1: 011001

word2: 100101

word3: 101010

Figure 6.1 of the paper

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